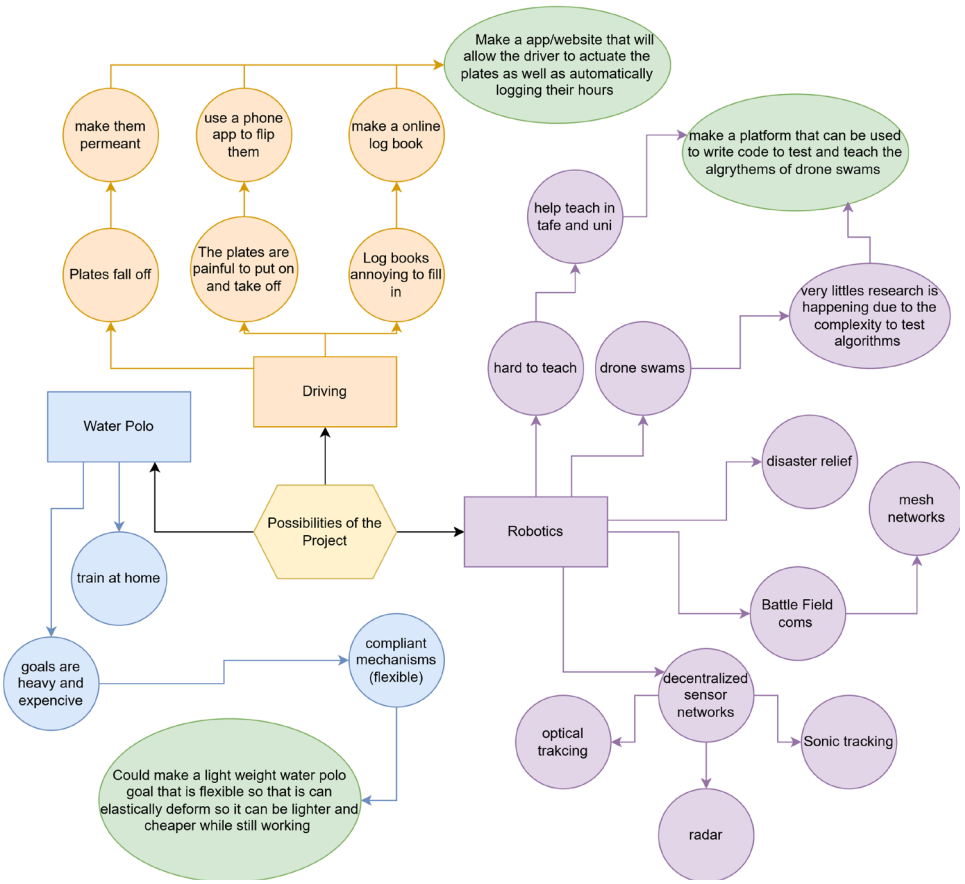
Project Proposal

Identification and Exploration of the Need: Exploration of Needs

Initial Mind Map: Several ideas were considered when selecting an option to pursue for my major design project.



Initial Ideas

Idea 1: The first idea was a drone swarm that can be used for teaching swarming technologies to students. The goal is to provide scientists and students with the hardware and software platform to develop new algorithms, as there is a lack of development in the field of swarms.



Ongoing Evaluation: The reason I did not choose this project was because it is incredibly complex and hence I would not be able to make a polished final project.

Idea 2: The next idea was to create a water polo goal, as currently water polo goals are very heavy and expensive. They are designed this way so when people shoot at the goal with great force it can withstand it. To solve this problem, I was going to make a goal that is flexible whereby the soft material could move on impact and then recover to its original shape.



The reason I did not choose this project was because it would be difficult to iterate and test new designs as each version would require a new goal, which is a significant time investment.

Idea 3: The final option and the project I decided to go with was an automatic L-plate. This L-plate solves the need that L-plates and P plates regularly fall off, leading to environmental pollution through the creation of litter and also great annoyance for the public. In addition, the ongoing cost of replacement. My solution would also solve the tedious process of putting on L-plates and taking them off as well as filling in the log book through an app I would build with it. People often drive with the wrong plates due to laziness, which is both illegal and dangerous. I have lots of experience making electronics projects and designing 3D printed parts but I want to learn how to optimise mass production through 3D printing as this is a useful skill for the future. This project will also help me learn about software design, app development, microcontrollers, and GPS.

The reason I did not choose this project was because it is incredibly complex and hence I would not be able to make a polished final project.

Project Proposal

Existing Products






It is difficult to determine existing products for this type of a proposal as there are none. The only functioning existing product that exists at the moment are physical L plates that need to be stuck on manually.

Why is it a need and what problem does it solve?

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

- Not filled out log books
- Sporadic driving practise → worse drivers
- Driving with incorrect plates
- Plates falling off → litter and annoyance
- Plastic Waste
- Inconvenience of removing and adding the plate to fit multiple drivers

Secondary Research

Plastic Waste

• Poor fixtures: L and P plates are required to be displayed outside of the car which can result in them getting damaged quicker. The clip/holder that are typically provided with the L and P plates are also generally made of poor quality plastic, adding to the waste.

• Non-Recycled Material: L plates create plastic waste due to the non-recyclable materials they use. So when they fall off on the street, they cause litter and pollution resulting in added issues to the enviornment especially if this happens in the bush and can not be cleaned up



This article...**summarise this article)

Website: https://www.drive.com.au/caradvice/this-growing-litter-problem-on-our-roads-has-an-illegal-solution/

This article...**summarise this article)



https://www.theguardian.com/environment/2025/aug/24/australias-amount-of-plastic-waste-surges-as-recycling-rates-fail-to-improve

Project Proposal

Secondary Research

Plastic Waste


Every NSW waterway contaminated with microplastics, study finds

By Kelly Fuller and Tony Ibrahim

ABC Illawarra | Water Pollution

Mon 16 Mar

Every waterway sampled contained microplastics, according to the report. (Supplied: EPA)

This article...**summarise this article)

https://www.abc.net.au/news/2026-03-16/nsw-estuaries-microplastics-report/106458862

This article...**summarise this article)


Ministerial media release

New research funded to find plastic waste solutions

Published: 5 February 2025

Released by: Minister for Environment and Heritage

Listen

Three pioneering projects have been awarded $1.25 million by the NSW Government to tackle plastic pollution through innovative and impactful solutions.

Previous governments left Greater Sydney on the brink of a waste crisis. Without new waste and recycling solutions, Greater Sydney's landfill capacity will be exhausted by 2030.

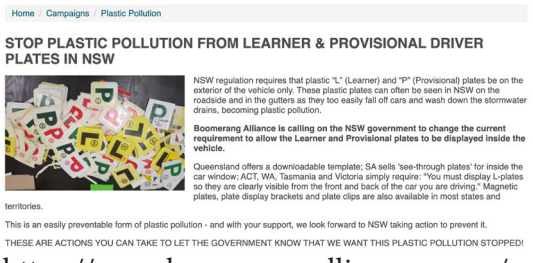
The Minns Labor Government is committed to solving the waste challenges and supporting future technologies that will continue to drive us to a circular economy where nothing is wasted.

Universities and government research institutions were invited to apply for funding under the Plastic Research Program.

Following a competitive process, three exciting projects were successful in securing funding:

https://www.nsw.gov.au/ministerial-releases/new-research-funded-to-find-plastic-waste-solutions

Boomerang Alliance


Home / Campaigns / Plastic Pollution

STOP PLASTIC POLLUTION FROM LEARNER & PROVISIONAL DRIVER PLATES IN NSW

NSW regulation requires that plastic "L" (Learner) and "P" (Provisional) plates be on the exterior of the vehicle only. These plastic plates can often be seen in NSW on the roadside and in the gutters as they too easily fall off cars and wash down the stormwater drains, becoming plastic pollution.

Boomerang Alliance is calling on the NSW government to change the current requirement to allow the Learner and Provisional plates to be displayed inside the vehicle.

Queensland offers a downloadable template; SA sells 'see-through plates' for inside the car window; ACT, WA, Tasmania and Victoria simply require: "You must display L-plates so they are clearly visible from the front and back of the car you are driving." Magnetic plates, plate display brackets and plate clips are also available in most states and territories.

This is an easily preventable form of plastic pollution - and with your support, we look forward to NSW taking action to prevent it.

THESE ARE ACTIONS YOU CAN TAKE TO LET THE GOVERNMENT KNOW THAT WE WANT THIS PLASTIC POLLUTION STOPPED!

This article...**summarise this article)

https://www.boomerangalliance.org.au/pplate

Identification and Exploration of the Need: Exploration of Needs




Although this is a good proposed solution, it still would not acknowledge the following issues

- Visibility: Many of the cars are tinted and it may also not be visible during the day due to the sunlight reflection
- Thermal: Cars can get very hot, up to 40-50 degrees celsius in the summer, which could result in the plastic melting
- Blocking vision for the drivers through rear window

How would my product improve user experience over existing or similar designs?

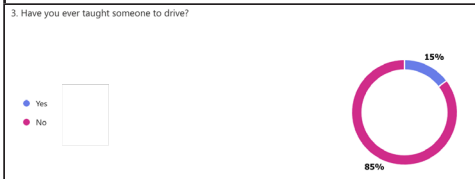
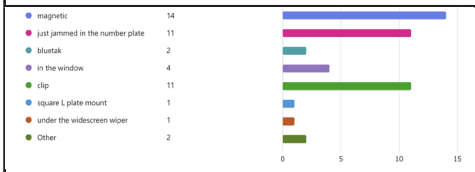
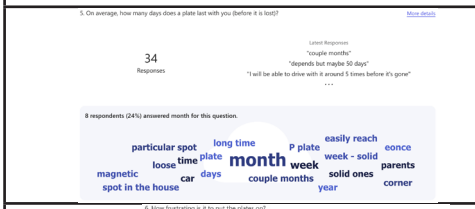
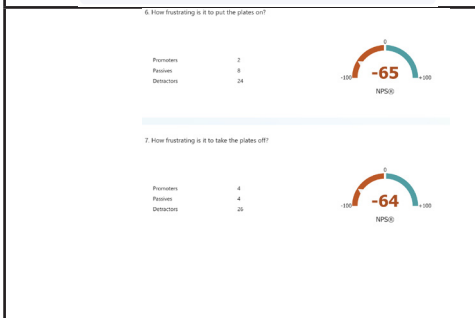
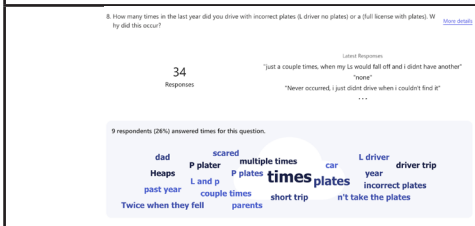
With fixed plates that can be electronically displayed and hidden my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Project Proposal

Point of Difference

Existing products are all a plate that is made of plastic or the black magnetic material that are attached to the vehicle by magnetism, a clip, suction, or adhesion (bluetick or tape). These mounting options all have their strengths and weaknesses but the best option is the clip because it is easier to take on and off and it is quite reliable. My design solution will stay permanently attached to the car and will use an electronic system to toggle the plate on and off. This electronic system will move a flap or similar mechanical system to controllably reveal and hide the L-plate, thus resulting in the effect of the plate becoming visible and not. This is quicker and easier than the manual systems. The app/website that I will build will allow the user to log hours easily and quickly.

Primary Research

	This defines the scope of the problem by illustrating the reality that these plates are lost fairly regularly hence the current system need, improvement.
	Demonstrates the log books are not filled in correctly. This shows that a solution is necessary to encourage people to fill in their logbook correctly complying with the legal requirements.
	shows that people falsify records out of laziness and other factors such as time constraints. This data provides evidence that people are not acting within the regulations of the laws, and then, consolidating the concept that users inherit laziness leading to danger.
	
	
	

Project Proposal

Design Brief

For my major design project, I plan to design an automatic L-plate. From my research and brainstorming, I have discovered a problem that I can solve which is the bad practise around displaying L-plates. Using my experience of CAD modelling and designing for 3D printing as well as my experience in electronics design and microcontrollers gained from personal projects and design technology, this give me an advantage when making this product. Using my skills, I intend to make an automatic L-plate this L-plate should allow the user to go from displaying them to not. This will be controlled form an app or a website on the user's phone a integrated log book that will be filled out automatically will be integrated into it.

Design Situation

The project is needed because L-plates are annoying to take on and off and they often fall off, causing litter. The increased convenience and better mounting design will result in fewer people breaking the law by driving without the proper plates, and improve the accurate filling in of the logbook.

Not filled out log books

Sporadic driving practise => worse drivers

Driving with incorrect plates

Plates falling off => litter and annoyance

How does it improve the user experience over existing or similar design solutions?

With fixed plates that can be electronically displayed and hidden my design removes the arduous and time consuming process of walking around to the front and back of the car to put L-plates on. Often learner drivers miss out on driving trips when passengers are rushed which reduces opportunities to develop their driving skills. The device will securely hold the plates to reduce likelihood of them falling off and littering the road.

Target Market

Non-fully licensed drivers and anyone who shares a car with them. This includes the L-plate divers leaning in parents cars and P-plate drivers who share with their parents.

Where do they live?

In every state of Australia.

Why will it appeal to the user?

This will appeal to the user because it is better than the other options in the following ways

Less litter from dropped plates and don't have to keep buying them

Integrated logbook filling out

Easier to switch from displaying plate to not (don't have to get out of car)

Helps you become a better driver because of the ai teacher built into the app

Project Proposal

Areas of Investigation



Areas of Investigation continued...

Area of Investigation	Justification	What will be investigated	How it will be investigated
Placement: Where can I legally put a L plate?			

Project Proposal

Areas of Investigation continued...

Area of Investigation	Justification	What will be investigated	How it will be investigated
Suitable Design - Size - Colour			
Mechanism - Motor input (app connectivity to the device)			

Project Proposal

Criteria to Evaluate Success

Criteria	Relevance	Implimination
Function - Ergonomics: Easy to take on and off - Quick to put on and off	The L Plate must flip open, shut and log hours fastrer than it is to do it manually. The time it takes a typical leaner driver to put my plates on is 46 seconds Displaying L-plates (and taking them down) must be significantly more convenient than the old way is because the user won't use it. Log hours must not take longer than manual and deliver additional benefits such as permanent record keeping (indelibility) The product will not be used, and therefore the need won't be solved	The L-plate should take 50 seconds to toggle and the log book component should take 48 seconds to complete I will be testing each prototype and the rate of which it takes to put on and off and also the ease of it. If there are any obstructions I will be improving this from prototype to prototype This also needs to be easy to use. So I will be evaluating the ease of the use of this product. I will also be surveying the demographic on
Aesthetics - Colour that does not stand out - Design should be sleek as it should go onto any car	The plate must blend into the cars aesthetic so that it is not an eyesore. f the plate looks ugly and does not blend in to the cars aesthetics, some users won't want to use it. It should be a pleasing design to look at and should not stand out too much on the vehicle. The impact if not achieved it the plate will have less adoption as well as not solve the need as well.This will result in people reverting back to the previous solution of the L and P plastic. The L-plate should not be visible foreign 10 meters away, and it should not have exposed wires. The colour should be neutral.	The L-plate should not be visible foreign 10 meters away, and it should not have exposed wires. The colour should be neutral.
Legality - Follows the NSW guidelines	Because this is a legally mandated signage, it must display the L-Plate in a way that complies with the regulations The entire MDP would fail because the plate would not comply with the legal requirements expected from the learner driver and their supervisor.	There is a minimum that is complying with the law or not it is black and white this is the linchpin of the whole project if it is not legal, the entire project is a failure I will follow the NSW and ACT guidelines and plate usage on vehicles closely referring to this website: https://www.service.nsw.gov.au/services/number-plates

Project Proposal

Criteria to Evaluate Success continued...

Criteria	Relevance	Implimination
Quality - Life Cycle - Can sustain wind and other road conditions	The plate should feel high-quality and be high quality. Because this is a 3D printed there is a mindset that all 3D printed products possess inferior quality. However, this is not always true. It is easier to make premium products if by taking full advantage of the 3d printing quality. If it does not feel high quality, people who care about their cars will not put it on the it. If the product is not high quality, it will not work well and break easily thus not solving the need of the plates falling off The impact of a inferior product will result in a smaller portion of the target market being addressed	There must be proper-looking packaging and no exposed wires The product must work reliably and have longevity. Testing will occur with the final prototypes of actually driving with the device attached to the car to ensure that it lasts. The material that the product is made of would also reflect a stronger material than the current solution.
Safety - Safe use for consumers - Should not damage the car	The product needs to be safe to both the consumer and the car itself. The consumers should not experience and cuts while attaching the product on the car The car itself should not get damaged when using the product. This includes paint scratches or dents onto the actual vehicle due to the product	I will test each prototype to ensure that it does not damage the car. A stress test will also take place to determine if there are any impact on both the consumer or the vehicle.
Cost	The L plate must be a reasonable cost. It should not cost than more units that one would spend at the moment. A physical pack of L plates typically costs between $3 and $15 AUD, depending on the attachment style. Basic Plastic / Tie-on Plates: Around $3 – $5 per pack.Magnetic Plates: Around $7 – $10 per pack. Clip-on / Premium Holders: Around $12 – $15 per pack	I will closely evaluate the cost of my experimentation and resources used throughout the process to ensure the product uses materials that do not exceed the concurrent cost.

Project Proposal

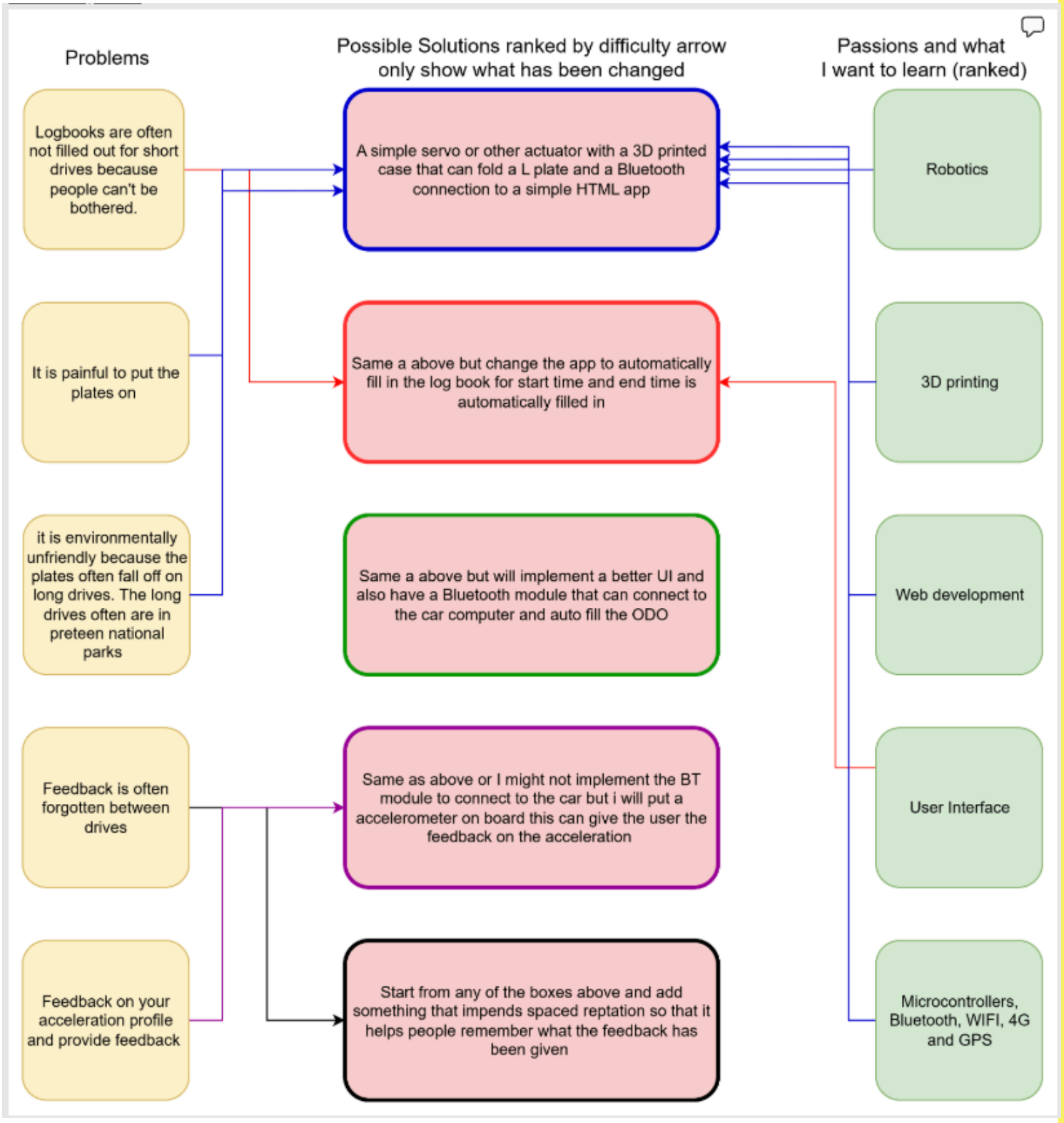
Time Plan

Project Proposal

Finance Plan

Item	Image	Proposed Cost	Actual Cost	Evaluation

Project Proposal
Action Plan



Project Proposal

Project Proposal

Project Development & Realisation

Existing Designs & Solutions

Image:	Plus:	Minus:	Interesting:

Image:	Plus:	Minus:	Interesting:

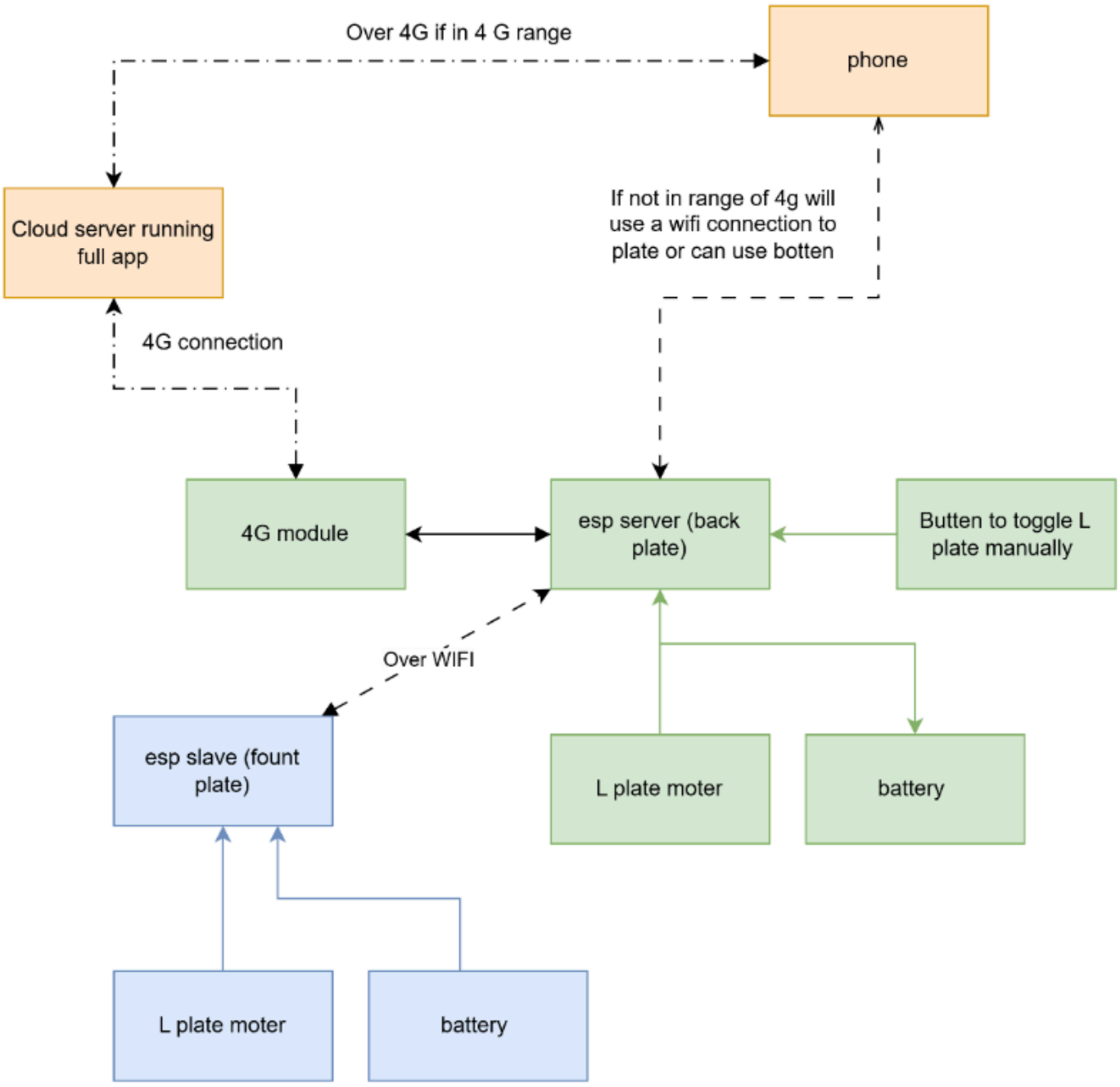
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Project Development & Realisation

Sketches

Project Development & Realisation

Idea Generation



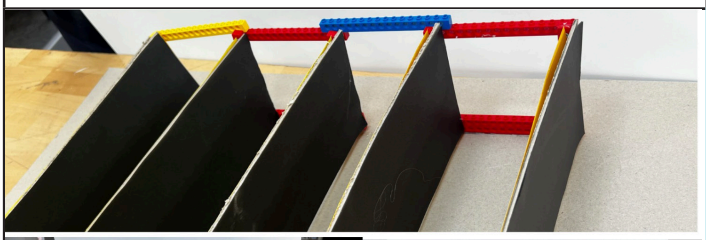
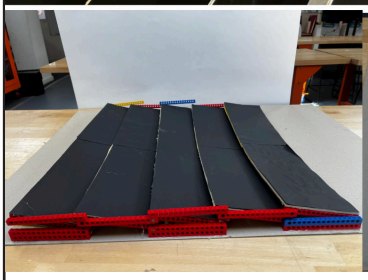
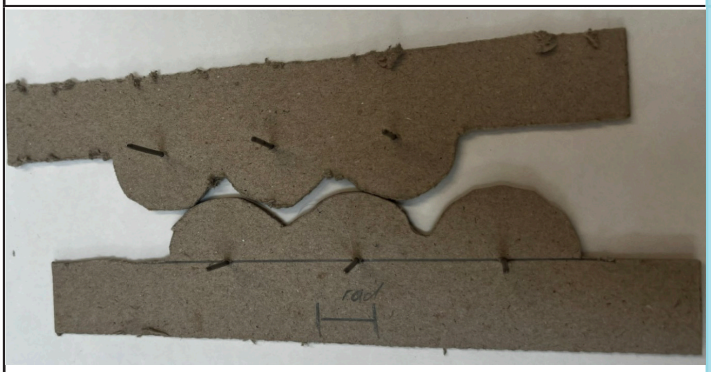
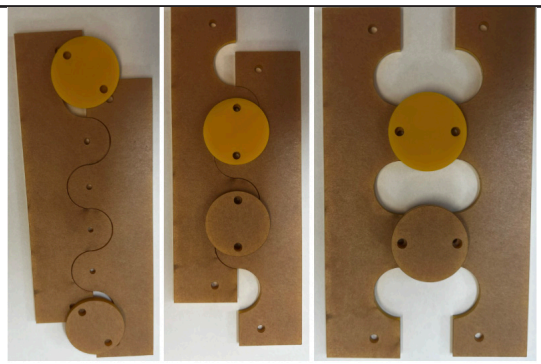
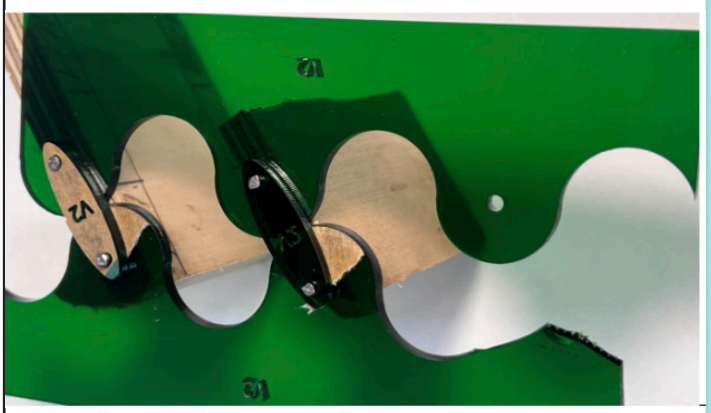
Project Development & Realisation

Factors Affecting Design

Factor	Evaluation

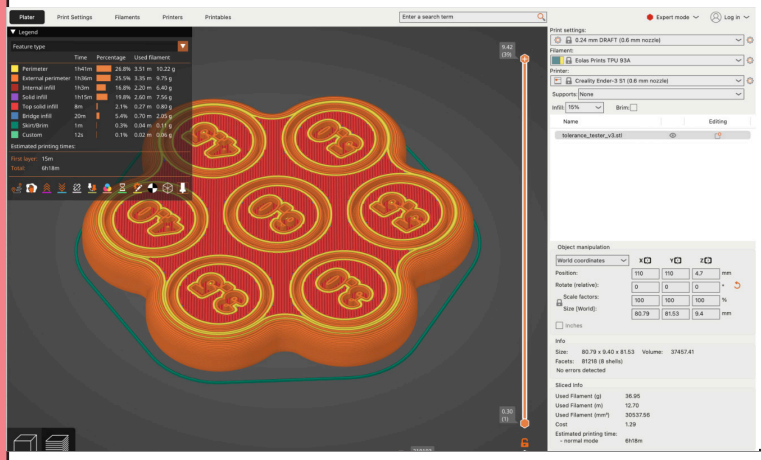
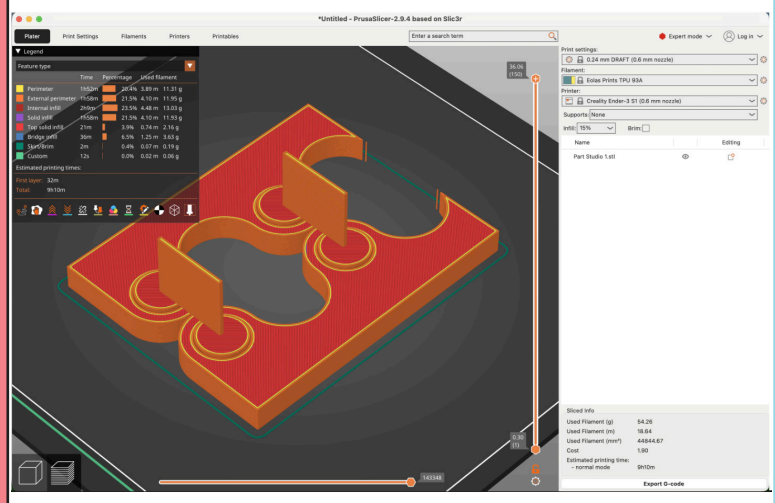
Project Development & Realisation

Prototypes

Prototype	Evaluation
  	Prototype 1: Template Steps 1. Used Lego blocks as a template to position the foundation blocks correctly 2. Hot glued one sheet of cardboard to the Lego bricks 3. Assembled the pins 4. Glued yellow and black paper to highlight how the mechanism moves and that it will work as an L plate This was the first test of the mechanism with the four bar linkage. This was an essential step to establish that this mechanism was a valid solution to the problem.
	Prototype 2: 1. I had a discussion with my mentor and we brainstormed ideas together until we came up with this idea of a mechanism 2. Next, I laid out cardboard and then cut it out 3. To simulate pins and joints I put nails through the centres of the holes 4. I then attached a plate to the nails to check if the mechanism could actuate fully
	Prototype 3:
	Prototype 4: I made the laser cut acrylic on Adobe illustrator and I adjusted the hole placement as per the previous one. I could reuse my same file but only making some small tweaks. I also changed from large circles to smaller ovals because this would allow me to see what was happening in the mechanism and better. This prototype taught me that my mechanism was completely valid although this particular mechanism was not optimal or optimised but it did work which was all it needed to do. There was lots of room for improvement but it was a foundation that would work.

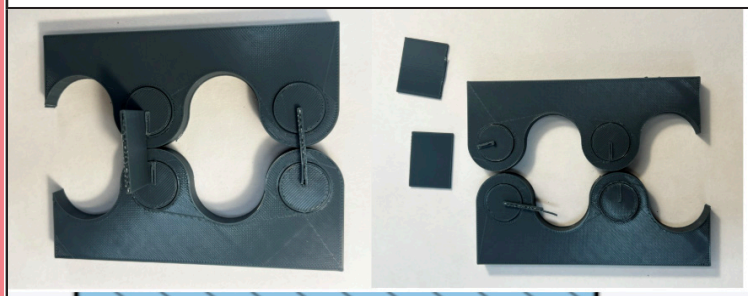
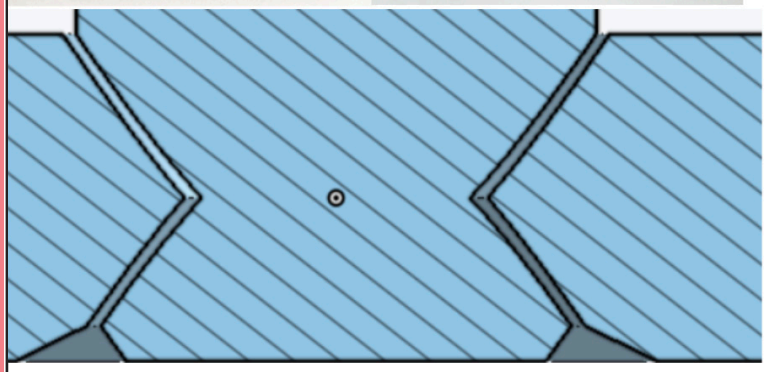
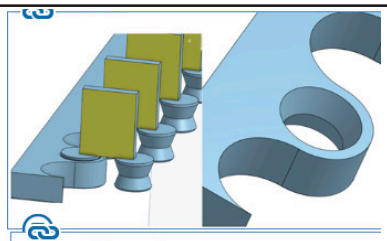
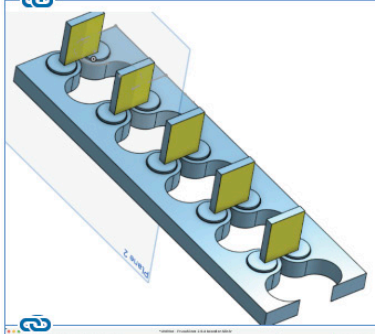
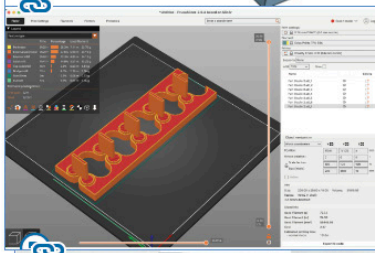
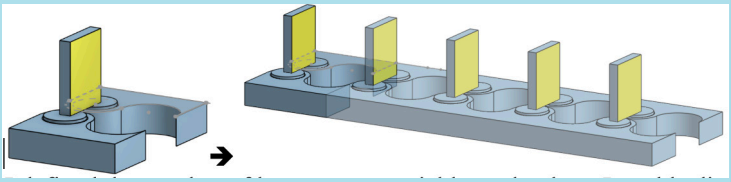
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 5: 1. Find the STL https://www.thingiverse.com/thing:3591280 it was on a site that is a Repository the 3D printed 2. Next I then imported into Prusa Slicer 3. I printed it on the school's 3D printers 4. The General process of putting on a 3D is to: 5. Load the filament 6. Preheat the bed and nozzle 7. Make sure the bed is clear 8. If necessary applying glue stick to the printer bed 9. Slice it with the default settings except a brim 10. Upload the sliced G code file onto a USB stick and begin the print 11. Click start 12. Monitor the first layer does not have any failures 13. Come back and take it off the bed 14. Then I took this off the printing bed and with my hands saw which ones of these I could move if I could move it it was a good enough tolerance if not it wasn't Ultimately the purpose of this was to be able to find out what is the tolerance of the 3D printer because I was doing print-in-place mechanisms and I need to know the gap that I need to leave so for my next prototype I can work out in CAD how thick it should be reducing the amount of iterations I have to do
	Prototype 6: I designed this particular part in Onshape I used a revolve to generate the dovetail pin I made sure I assigned the particular tolerance to a variable in the sketch I used so that I could easily modify it without needing to edit my sketch manually this is called parametric design and I used it throughout the entire design cycle exported the 3D model and then I printed it using the standard process

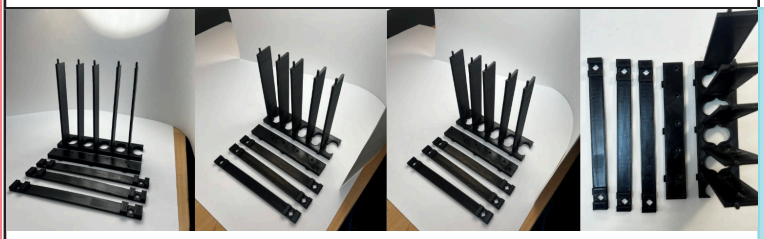
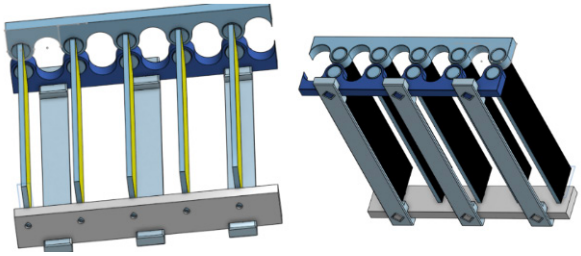
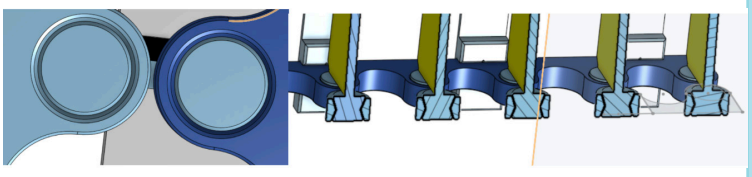
Project Development & Realisation

Prototypes

Prototype	Evaluation
 	Prototype 6 continued.. this allowed me to verify that the dovetail mechanism that I'm designing works as well as confirming that the design and shape work that was also additionally confirmed in the green acrylic 1. This is very useful because it allows me to as a small scale prototype refine the mechanism further knowing I have a working mechanism that only needs to be optimised from this point forward.
   	1. Given that you can quite clearly see a repeating pattern in the design that I wanted to achieve I was able to leverage the ability of CAD to do patterns so I started by making a single copy of what I wanted and then I duplicated it the number of times I wanted. 2. I defined the number of louvres as a variable so that later I could adjust the number of louvres I needed. This was very useful when testing 3, 4 and 5 different numbers of louvres.  3. I then printed it using the standard procedure Once I had printed this and verified that it worked, I could also test that a single one of these louvres could drive the entire mechanism. This allowed me to confirm that a single driving motor could work, which is the most efficient way of driving it. This is very beneficial because it confirms not only the mechanism but the way of activating it as well in future. I can start designing and optimising this mechanism for mass and area coverage as well as how to mount and drive it with a motor.

Project Development & Realisation

Prototypes

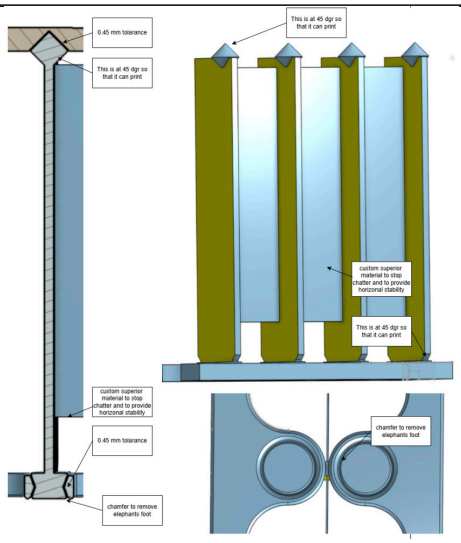
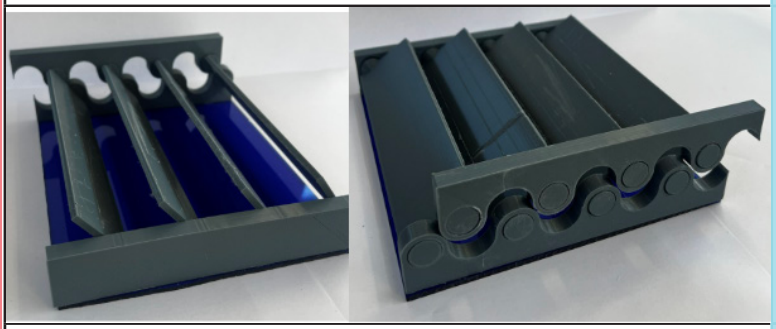
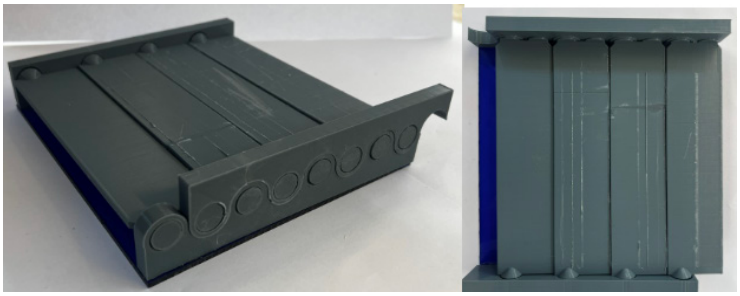
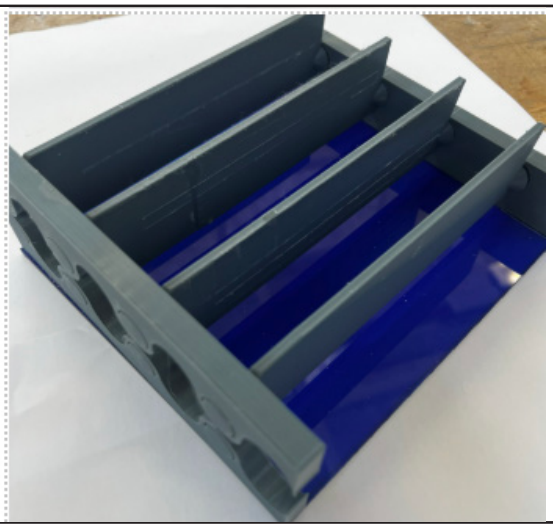
Prototype	Evaluation
	
	This image highlights the limitations due to chatter of the top pin
	These 3 images show the chatter this is why I added the custom support materials
	An overview of the mechanism
	
	Two images of the tolerances and the print in place mechanism

1. I was able to rapidly design to test whether this would work by not following parametric modelling procedures but instead just making it
2. The benefits of this is that it was faster but at the cost of having a reusable CAD model but I was able to revert at the end of this iteration back to the original model
3. One thing I did particularly was I made teardrop shaped holes. This allowed me to print at a better orientation as a teardrop hole will maintain tolerance as it won't sag
4. I also printed this using the standard procedure

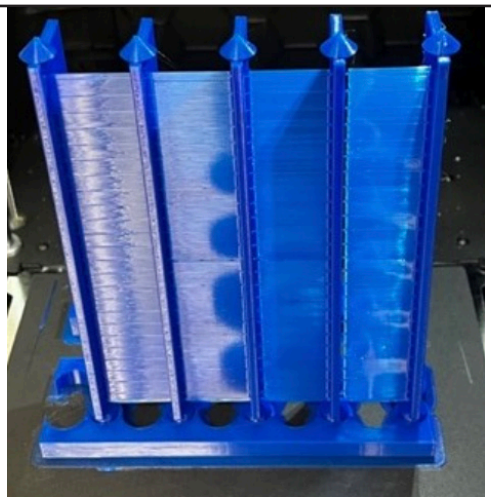
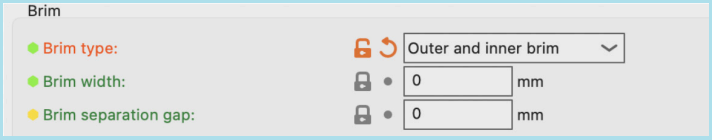
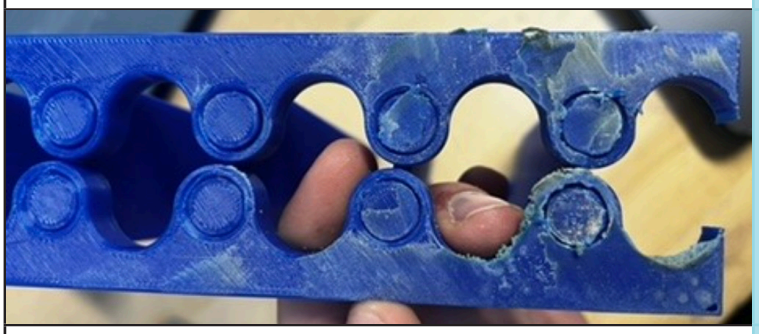
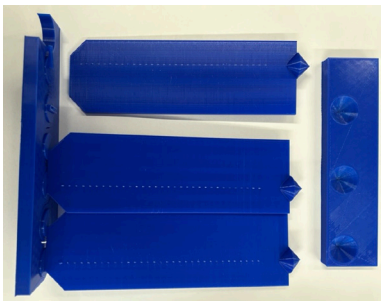
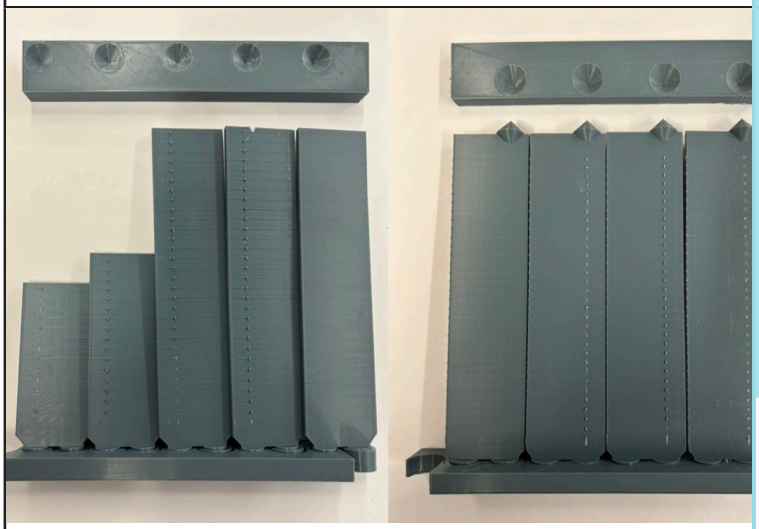
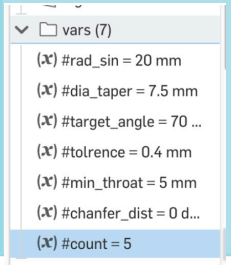
Ultimately this iteration of the prototype was somewhat unsuccessful but it did prove that the mechanism can work with an endpoint and full length levers but the downside was that the pins are really small and the bottom mounting method was not very reliable or strong as well as blocking vision so in future I need to work out some solution like a clear backing to join them as then it would not block vision

Project Development & Realisation

Prototypes

Prototype	Evaluation
	1. After researching about 3D printing optimisation I applied lots of the things I learnt as you can see in this annotation. 2. For example I added chamfers to remove elephant's foot. This is where the print squishes out and would cause the joint to adhere. I also added this large support material to stop them swaying. This is called design support. 3. The benefit of design support is if eventually I was to get this manufactured commercially. I wouldn't have to tune slicer settings on their end but I could simply get them to slice my model and it would work. 4. The reason I added 45° chamfers at the end is as for locating as well as maximising surface finish
	
	
	This print was incredibly instructive as it allowed me to determine that a lot of my optimisations were better as the mechanism that was less binding than the previous iteration as well as printing much more cleanly although the support material was very strongly adhered to so in the next iteration I should add preparations to allow it to easily detach. Despite me adding a chamfer for elephant's foot it was not large enough to properly mitigate it as I still had some evidence of elephant's foot so in the next prototype I will increase the amount of elephant's foot prevention I do

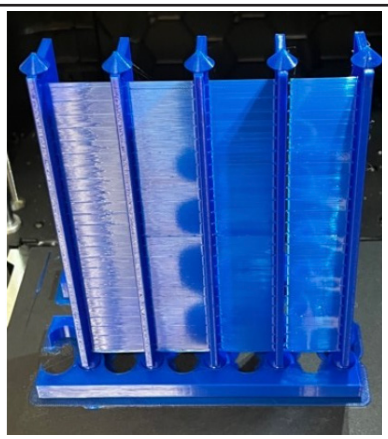
Project Development & Realisation

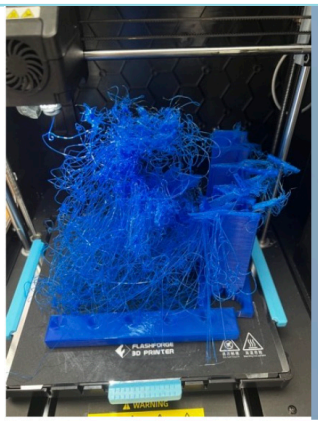
Prototype	Evaluation
	In this iteration I decided to add a brim. This would enable me to print it with better support and it did work, but it was almost impossible to remove and it damaged the mechanism. 1. I in CAD added perforations in the support material 2. In the slicer, I toggled the option for brim  3. As a side-effect of this I had the mechanism be bound up so to fix this I used a disc sander to grind away at the mechanism where it would not come away
	This wasn't good because it left damage to the mechanism, but the mechanism did function Going forward I will know that if I use a brim I should select an outer-only brim so it is easy to remove, otherwise I will struggle removing the brim.
	
	1. By adjusting the parametric variable for the count, I was able to generate three different louvres with different numbers of blades 2. 3, 4 and 5 successively this let me test what was the optimal number of blades for my prototype 3. I printed each of them using the standard procedure. I printed them all at the same time, though with different filaments. 

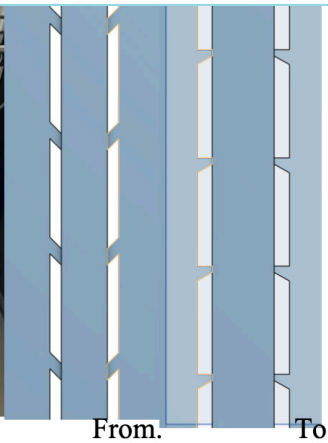
This let me determine the optimal number of blades to have on my mechanism as it was quite difficult deciding on the optimal number. As it is a compromise between the smaller I make it the smaller the pins have to be that are printed so I would like them to be as large as possible but then I've a side-effect of having to have thicker circles and larger blades which take up more space so to satisfy my design criteria I had to choose an optimum of these two that is why I decided at five as it seemed to satisfy all my requirements

Project Development & Realisation

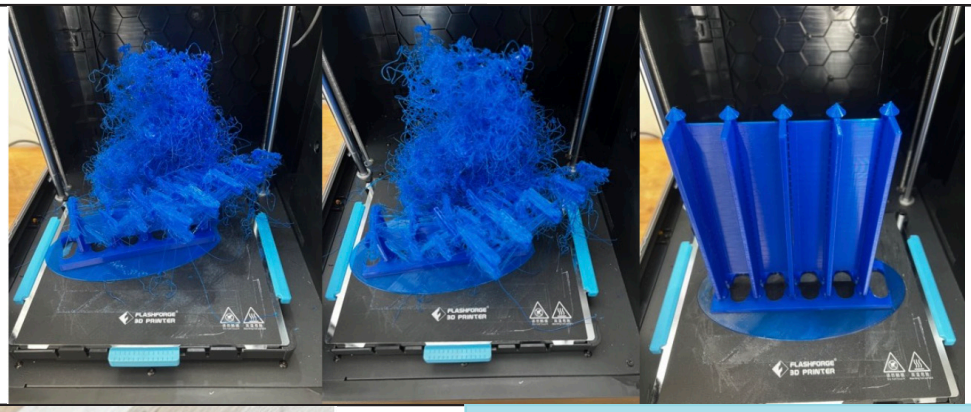
Prototype	Evaluation

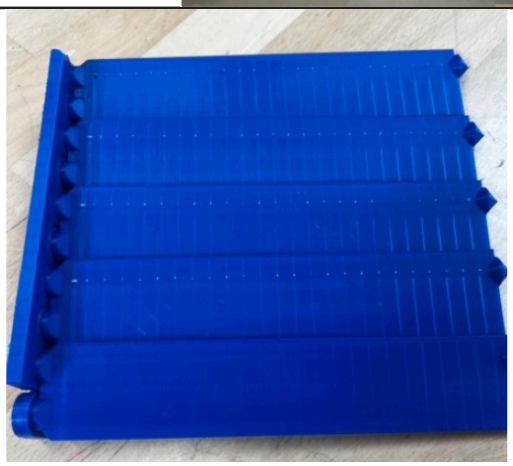






From. To





1. Onshape I changed the thickness the designed support from 1.414 mm to 0.400 mm so that it would be easier to remove I did this by editing a sketch

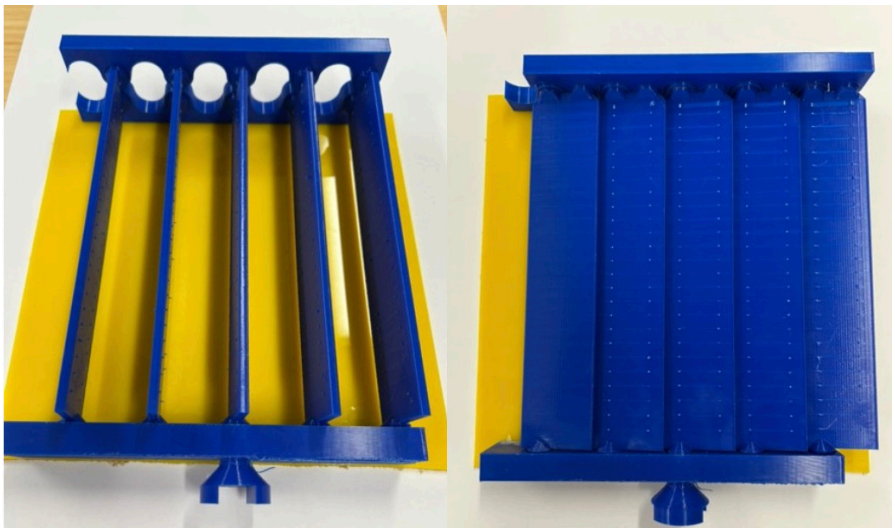
2. Then I printed it with a brim as seen in figure 1 this made it difficult for me to work out what my changes were as I forgot that I had made the small change in the thickness of the support material

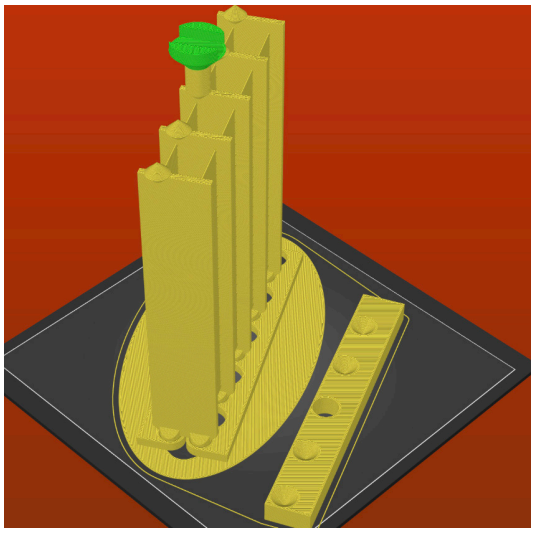
Basically what I did is I first printed one of them and then I printed it again with a better layer height and I did lots of very good quality bed prep with like glue stick and that also failed so then I printed it with a designed brim and that also didn't work. The issue was I kept on getting layer shifts and I didn't know why and it was actually because the louvres were existing as I can't believe her and they were snapping their limited area of the bed the printer so this fixed by me adding support material so that they couldn't leave it and it was actually one homogeneous piece so it would have enough surface area to stick to the bed and I did this across multiple iterations and I also added a brim in this step. This is a printing place support material which increases adhesion. I don't think it's necessary anymore like for the actual final one but I've kept it in there because it looks good, like I've designed things and it's complex.

This taught me that despite having thin support material which makes them very easy to remove You have to make a compromise about too strong or not strong enough and I made the wrong compromise by making it so thin that it came off and it didn't actually act as support material. That's why these prints kept on failing across three versions.

Project Development & Realisation

Prototype	Evaluation





1. To make this I didn't have to modify much. I modelled the actuator mount itself in CAD, and the only physical change I made was drilling a 13 mm hole for it to fit through. I used a drill rather than remodelling and reprinting the whole part because I already had one printed, so this was much faster

2. Then what I did was I pushed the print midway through. You can do this in the configuration in the slicer settings so that the print will automatically pause at the correct layer height. You can see this in the image with the green and yellow a slicer screen shot. The green is where it all pauses so you can change colours.

3. This allowed me to place the bar on top of the louvres in place and then I could resume it and print the drive mechanism

4. Once I finished the print, I took it off and glued it to a base and I also lubricated the hinges

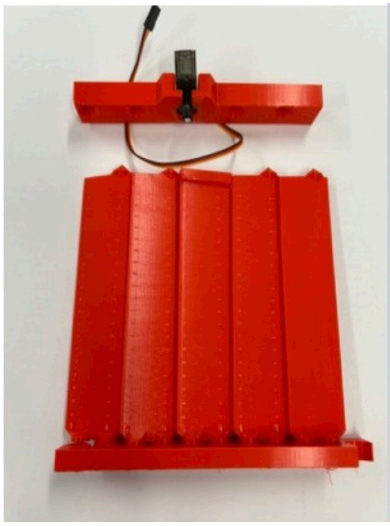
This let me establish whether it was possible to drive this mechanism from one single point as this is how I intended to design and make a servo mechanism for it to go into. I could also do load testing with the servo to see if the servo was powerful enough to turn it and it was, which means I can use a small SG90 (this is a typical small servo driven by a PWM [pulse width modulation which means a series of pulses which encode digitally the position that the closed loop controller must try and drive to] signal from a microcontroller). Having a servo means that I can be more reliable and precise as a servo is a closed loop controller which means it knows position so if the gears slip or a kid holds it down it will still try and move once pressure is released instead of losing position as the case with 3D printing stepper motors.

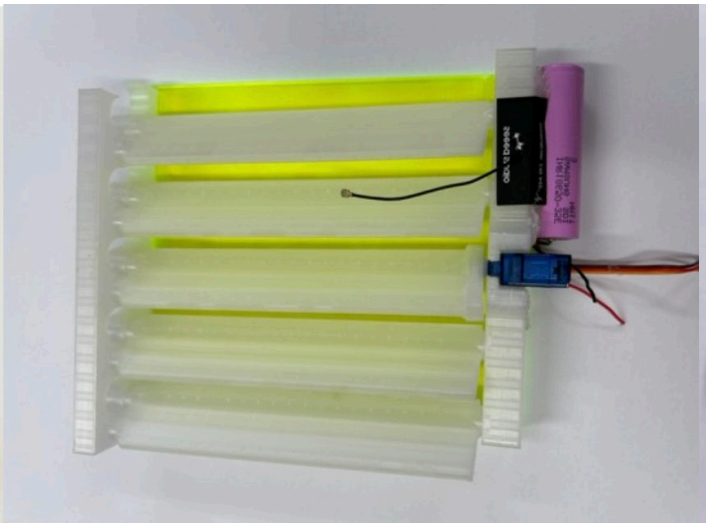
In addition this pausing the print and resuming is how I'm going to have the multicoloured L plates and it was a good verification of this manufacturing technique

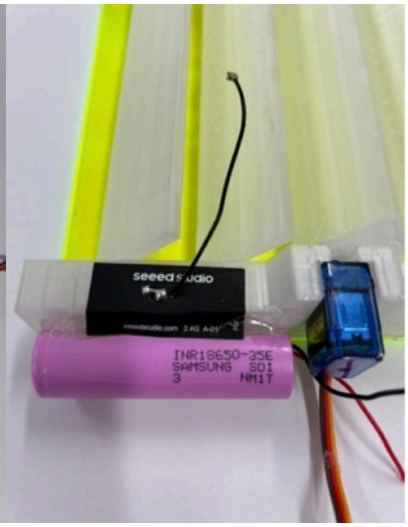
At this point, I'm quite happy with a louvre mechanism so far and I would like to move onto now prototyping the electronic platform

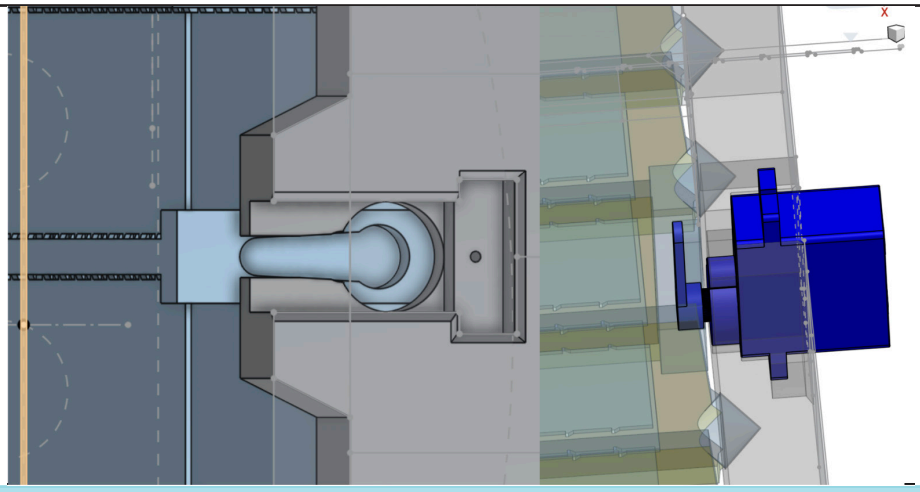
Project Development & Realisation

Prototype	Evaluation









1. The first step whenever you're modelling something in CAD to work around an existing part is to always import or make a model of the object you're trying to make a mount for or at least a rough approximation of the parts that you need to interface with

2. This meant that I imported a 3D model of the SG90 servo with a horn that I was planning on using as you can see in the image.

3. Then using a series of extrudes and sketches I cut away the material. In these sketches. I included an offset with my tolerance dimension. This is a globally defined dimension which is set to 0.25 mm. This is the tolerance of the 3D printer and it means that when I insert things it'll have enough clearance to physically fit.

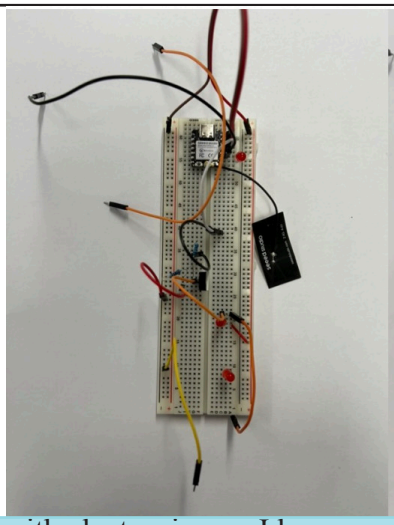
4. I decided the second servo arm mount is not necessary so I will be cutting that off so it can fit and reduce space which is one of my success criteria

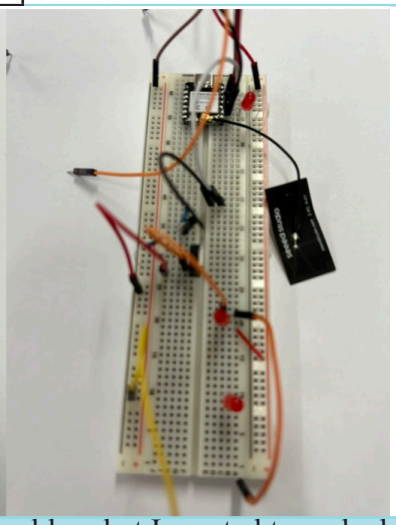
The red one Is printed in PLA whereas the white one is printed in PETG. PETG is known for better UV resistance and also being tougher. While PLA is much faster to print and good for rapid prototyping it does not handle UV well so that is why I printed in PETG so that I could get a feel for what that material is like. Going forward I will continue to use PETG when necessary but if there is no PETG in the DT building I will use PLA as it's fine for a mechanical prototype but does not necessarily show the final product or the same printing tolerances.

I've also learnt that my servo is strong enough to drive the mechanism. This means going forward that I can use a servo and continue designing and pursuing this design route. Next. I need to work out some way of mounting the battery which is not just with hot glue as currently it's crude.

Project Development & Realisation

Prototype	Evaluation

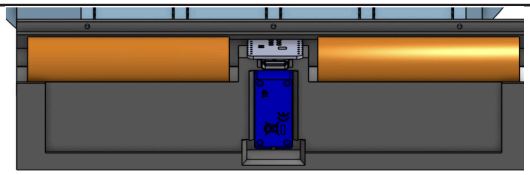


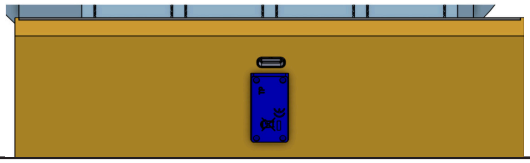


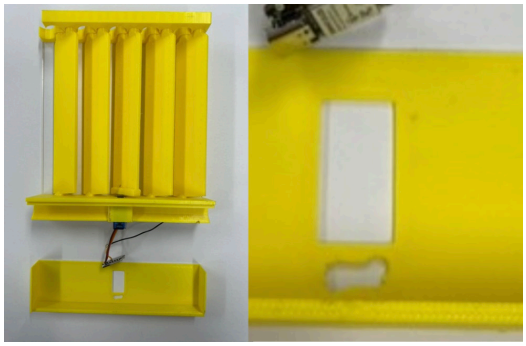
I have had some experience with electronics, so I knew roughly what I wanted to make but I was in contact with my mentor and ended up having a two hour conversation where we planned out the electronics of the board. This was our first version and prototype. It worked, but I decided not to keep the mosfet (it is essentially an electronically controlled switch, this particular mosfet defaulted to off and only turned on when a voltage was applied to it). The idea was that it would let me completely disconnect the servo so it could not draw any power at all, and I could also put the ESP32 into deep sleep, which would have given me much better battery life. In the end I decided it was not worth it, as the mosfet added cost and complexity to the board for a fairly small gain.

But instead I have decided to go down a route where I will just send a PWM signal of nothing to the servo which means it will not try and actively drive and finish the control loop which will have 95% of the battery savings while reducing weight and cost the end user.

Prototype	Evaluation



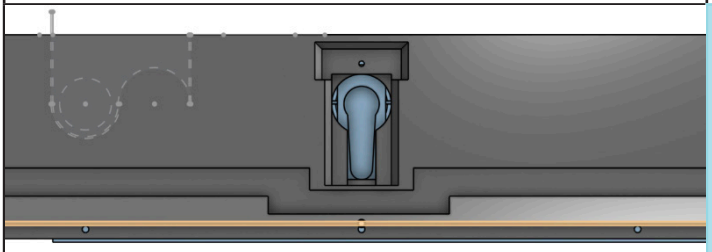
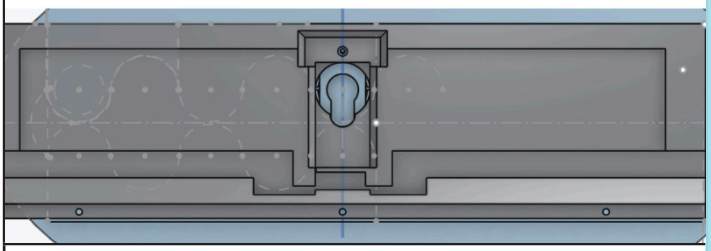
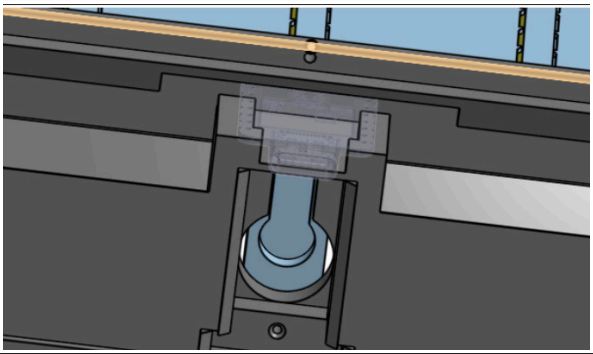
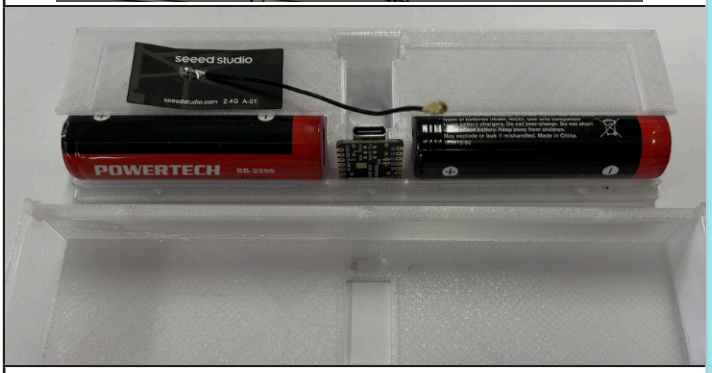
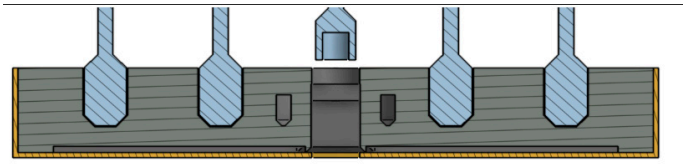
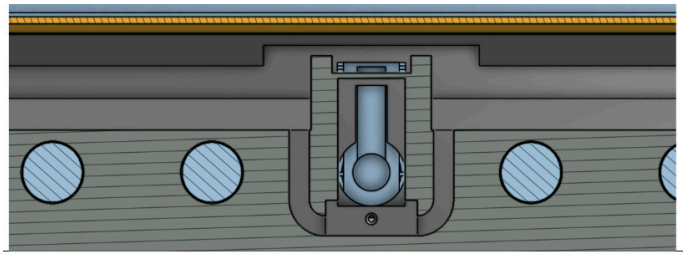




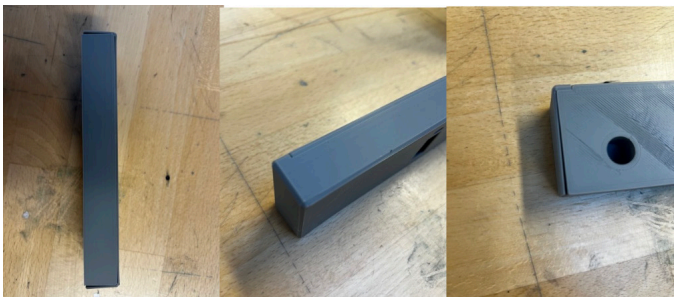
1. Following the established design pattern of placing the components first and then modelling around them I placed the batteries in a spot that I liked
2. To mount the batteries I made a V groove. This was a easy model and there was room to route the wire underneath and see it was also had a snap fit for the batteries would stay in place.
3. For mounting the ESP32 I had made a decision to mount it with the PCB labelled down. This allows the antenna exposed and the USB cable to be accessible but because the mounting was not repeatable and because of the heat generated this led to the ESP32C3 (this is a low power micro controller with Bluetooth and Wi-Fi functionality and an integrated battery charging module)

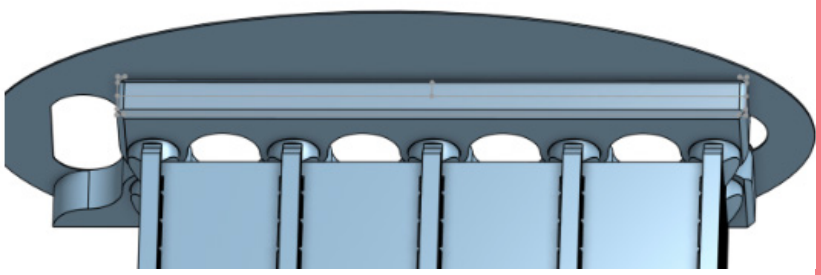
Things I didn't consider and should have : I didn't consider the wire routing as wires have a physical size and I didn't consider how they should be routed and this meant that the case would not fit on. I also did not make the hole correct spot or the spot was correct for the CAD model but since the ESP32 could move it didn't end up being in the correct spot so I had to use a drill to try and widen it which did not work out very well. I also didn't consider that there was no way to access the charging module pins underneath the ESP32 as they were hidden by the plastic. These various mistakes are what happens when you're on the first or second iteration of your electronics module and there's a significant amount of moving parts and complexities that you have to keep in your head and you make mistakes. That's okay because this is why we prototype. I needed to make sure that I include proper wire routing (this is foreshadowing) and also mount the servo correctly

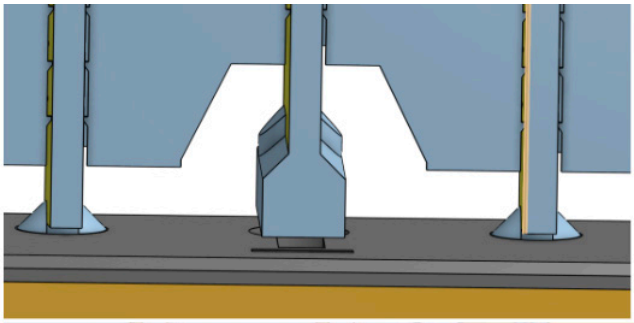
Project Development & Realisation

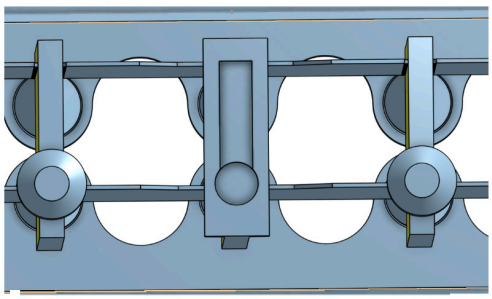
Prototype	Evaluation
   	1. As wires are very difficult to model I also made an error in this one. I accounted for the wires on the battery holder but I didn't account for the thickness of the wires going from the servo motor to the ESP32. I also made fix mounting on the top I did this by turning the ESP upside down and then doing a series of sketches and extrudes to make this boss that the ESP could sit on the back of the ESP was constrained by the back plate and the sides around the CPU and the ESP and the USB port were constrained by the two walls meaning that it would be in the same spot every time. I also adjusted the location on the outside case accordingly. 2. This let me make the ESP be in a repeatable spot every time which allowed me to design a correct size hole Going forward, I need to make a solution for routing the wires correctly This new layout worked really well although there was an issue with there being no way to access the boot button on the ESP32 the boot button is needed to reset the ESP32 when programming it with code. But other than that this setup was very effective despite that being no way of routing wires from the servo to the ESP32 that did not the case to bulge. You can see in the photo above that I have used a drill to test a different way of routing the wires. My first approach did not work very well but my second approach routing them through the body worked very well. That is what I decided on going with for the final wire routing solution from the servo to the ESP32
 	The reason that the channel on the servo wire routing Is curved is the orientation that print out and it's best to have a 45° angle. This is what I learnt in my research and a 45° angle or a horizontal 90° angle that is the optimal because if it is a horizontal, the printer can't stretch the filament from one pole to the other like a bridge and if it is 45° it is a general rule of thumb number about how steep of an angle it can have before it starts sagging so I used a combination of 45° angles and a bridge, which is the name for the 90° horizontal. This let me effectively route wires through a channel without the risk of having filament in the way as it would be difficult to clear out. I chose to make the channel on both sides instead of one as there was no reason not to if I make it on one side and it turns out that for some reason the batteries sit better if the wire goes the other way then I'm in a bad issue, but if I have it in both ways, not only makes it stronger because there's more material there because remember infill is not as strong as but that it will sit better depending on the particular case

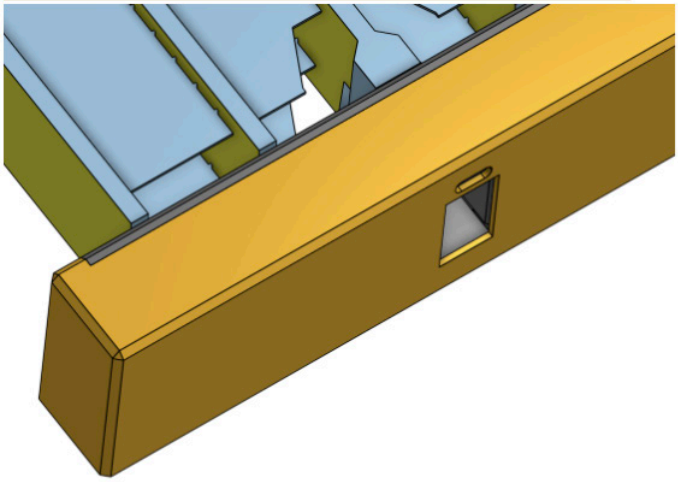
Project Development & Realisation

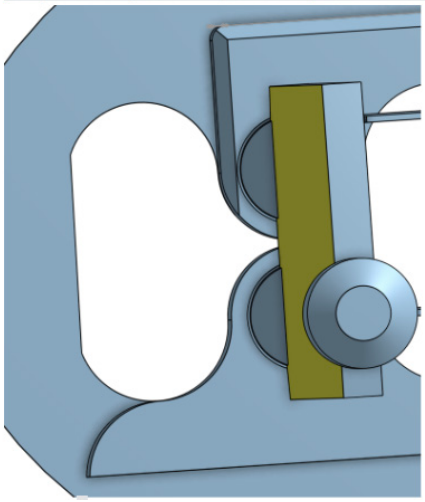


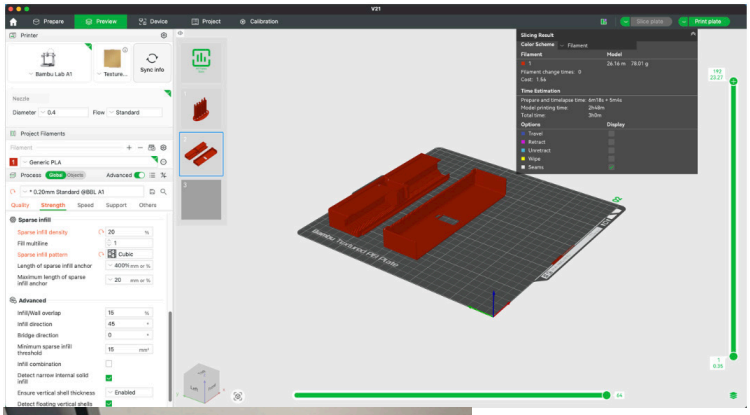


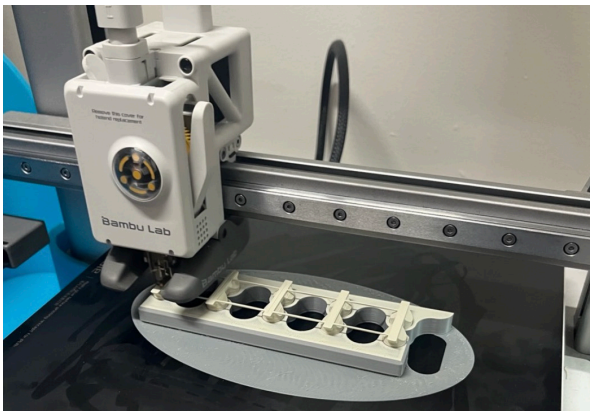


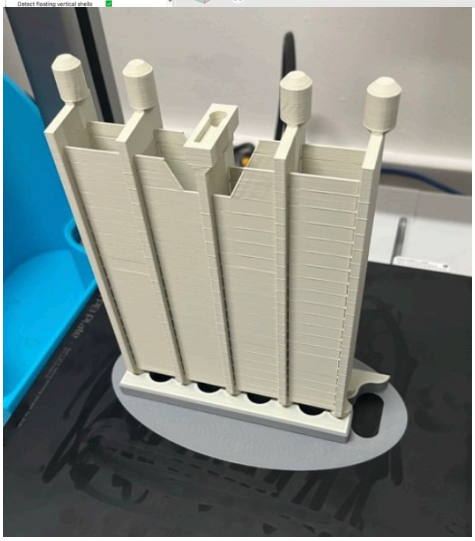


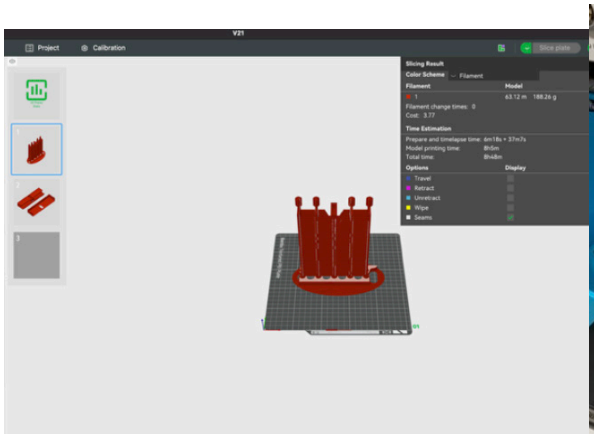


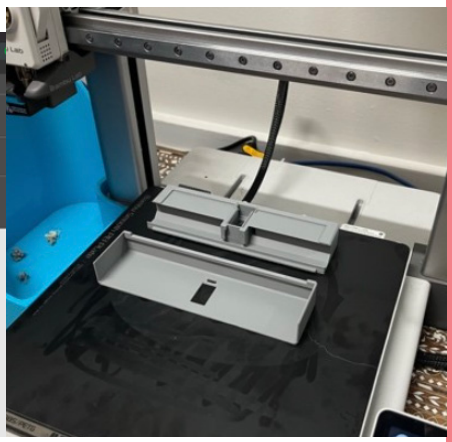






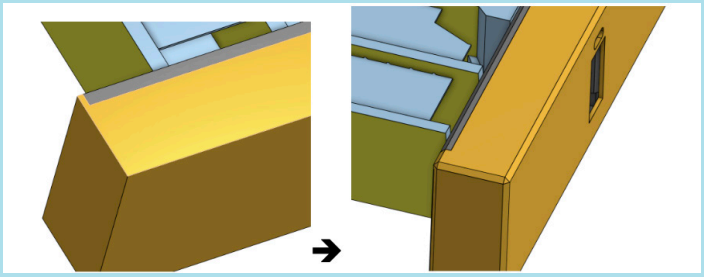






Project Development & Realisation

1. I wanted To increase the hole where the servo horn mounts as in previous iteration I had to use a drill and a heat gun to mount the servo horn into the hole so to do this I used an extrude but I also needed to chamfer for the sides of the Horn mount as it would mean that it would be stronger and not have a chance to fail as it's printing in thin air as a previous iterations did
2. I needed to remove material from my support material so that it could accommodate the larger horn. I did this with a simple sketch and an extrude the whole way through.
3. I wanted to make the edge of chamfered to improve the safety of my product to do this I needed to often chamfer manually as I wanted to keep the brim place.
4. You can see highlighted and figure 2 that the brim looks like it's in place and weirdly fitting this is because instead of properly modifying the parametric design I simply used an extrude to cut away and adjust the shape. This is risky but it is not worth modifying the original brim as there is a small cost in filament but a large cost wasted time fixing the parametric document so it would be easier just to cut away.
5. Because the corners were originally mounting in the edge



7. You can see how I move the location of the corner to be inwards so that there would be a solid material to make the chamfer on otherwise you would end up with a gap as a tolerances on chamfers are slightly worse than normal and this would look ugly and could cut someone but because I moved it inwards now you have a solid piece of material you're chamfering which both increases its strength and also improves the aesthetics

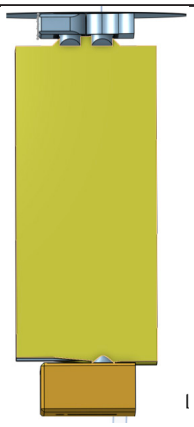
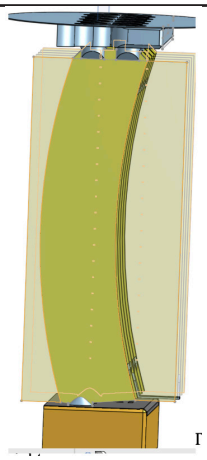
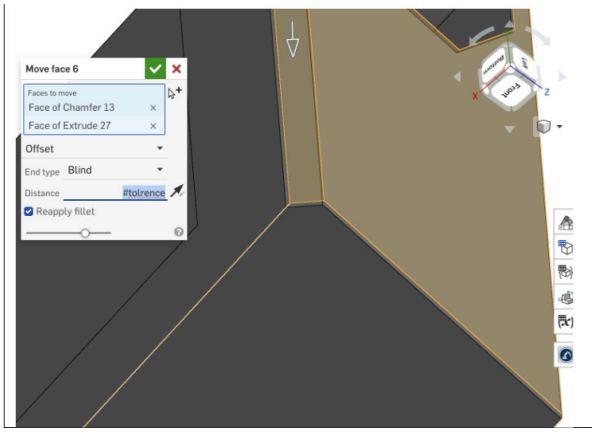




8. You can see this top corner. I did not include any tolerance. This means that the top corner acts as a focal point and it will pry open the bottom of this bar. Next time or for the next iteration. I need to increase the chamfer amount on the inside as I do on the outside so that it is a relief so that it doesn't pry open the lip

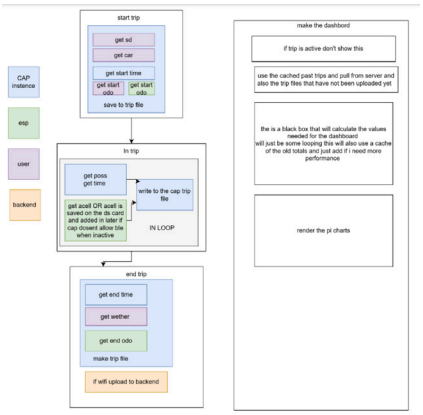
Overall the chamfers look really nice and clean but there needs to be improvement with refining tolerance in the case but ultimately these are small refinements that make a product high-quality and good design and it's important to get this right.

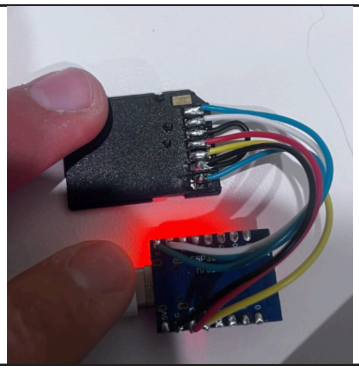
Project Development & Realisation

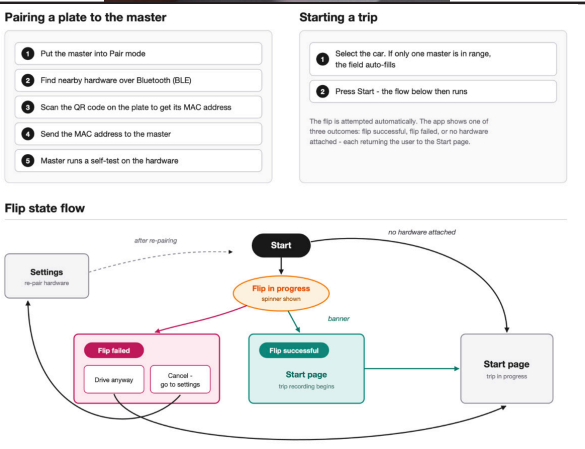
Prototype	Evaluation
	I extruded this to be like this then I can cut them to make it curved
 	Then I used a sketch and a extrude to cut away the luvers in the right way Then I used a file to remove the gap I had to remove material that was both in the shampoo and on the side face so once I've done that I removed that material in CAD for the next iteration

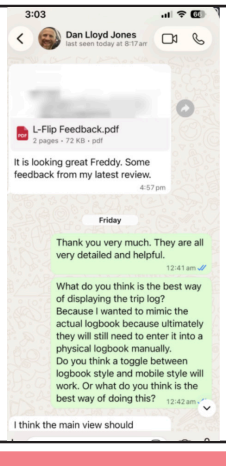
Project Development & Realisation

App Development









Communicating with a Mentor he is a professional software developer of mobile apps

Project Development & Realisation

App Development

Acceptance testing L Flip app . page 1 of 3 — 8 passed . 4 failed . 13 cases

#	Test case	Intended result	Actual result	Result	Action taken
1	**Register a account**	Taken to dashboard	Taken to dashboard you were routed through the login screen but this is so that I don't have to ...	PASS	NA
2	**Sign out**	Taken to long in screen without a jwt or cookie	I got taken to the onboarding screen	FAIL	Remove the cookie
3	**Bad password Bad email Bad name**	Rejected by a generic but helpful error message	I got good error messages that did not give away the back end infrastructure but were clear enough to ...	PASS	NA
4	**Test ai tool use**	Returns personalised information about you dives	I got a "Sorry, I couldn't produce an answer for that. Please try again." I got this message	PASS	the reason that this did not work the first time is because I made the chat and then ...
5	**Test that memory data is removed on log out**	Verify that the last ai chats are gone	They were not removed	FAIL	Execute a force reload this is a easy fix. Open father Investigation there a other ...

Wording shortened to fit. Full text in acceptance_testing_FULL_TEXT.csv — Documentation section 4.1

Acceptance testing L Flip app . page 2 of 3 — 8 passed . 4 failed . 13 cases

#	Test case	Intended result	Actual result	Result	Action taken
6	**End odo validation**	Verify that there is some sought of error message that don't allow this	Got a helpful error message	PASS	na
7	**Test log in**	You log in Successfully	It worked but I got the onboarding pages This will be annoying	PASS	Make the onboarding only come of if flag exists not a if flag doesn't exist Fix in ...
8	**Verify dark mode works**	I would expect that all the text is readable and the themes match up with no excessive amounts of light ...	I found a few text boxes and themes that were not properly in dark mode	FAIL	update my CSS and theming as well as my colour palette to work with dark mode
9	**Verify AI data filtering works**	The expected result is in the first case to get trip information and other specific information about ...	My server hardening worked verry well. But the ai tool fabricated data see 1) for the output this is ...	PASS	The fix is to harden the system prompt

Wording shortened to fit. Full text in acceptance_testing_FULL_TEXT.csv — Documentation section 4.1

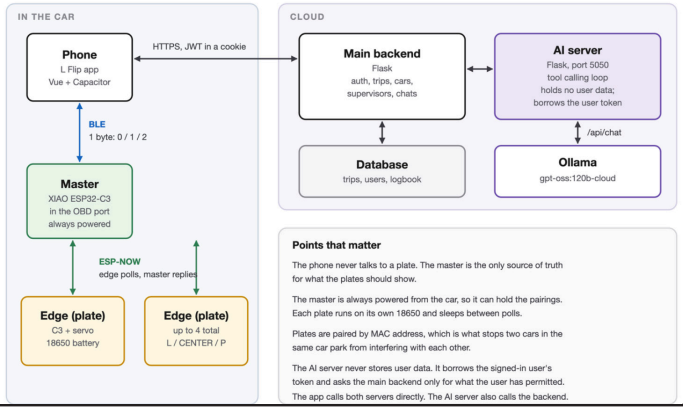
Acceptance testing L Flip app . page 3 of 3 — 8 passed . 4 failed . 13 cases

#	Test case	Intended result	Actual result	Result	Action taken
10	**Test full Creating a trip with screen off**	I'll be able to see it in the GPS map. GPS points will still be logged despite when the screen is off ...		not run	
11	**Testing the token limit**	Get a pop-up or other notification from the LM that you're out of tokens	This was really bad design on my behalf All you got was a pop-up saying please check your connection ...	PASS	I was HTML escaping the content twice which meant that I was getting things like " ...
12	**Verify hardware pairing works**	The master and slave will connect. Then they will seamlessly toggle with an intuitive UI	This works as expected	PASS	NA
13	**App icon look good**	The app icon should be a bright yellow with a black L	The app was a olive green yellow. This looked bad.	FAIL	So I changed it to a bright yellow

Wording shortened to fit. Full text in acceptance_testing_FULL_TEXT.csv — Documentation section 4.1

L Flip system architecture

How the phone, the cloud and the plates fit together.



Points that matter

The phone never talks to a plate. The master is the only source of truth for what the plates should show.

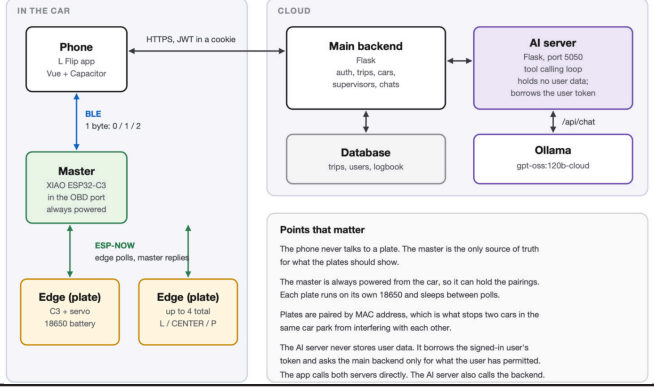
The master is always powered from the car, so it can hold the pairings. Each plate runs on its own 18650 and sleeps between polls.

Plates are paired by MAC address, which is what stops two cars in the same car park from interfering with each other.

The AI server never stores user data. It borrows the signed-in user's token and asks the main backend only for what the user has permitted. The app calls both servers directly. The AI server also calls the backend.

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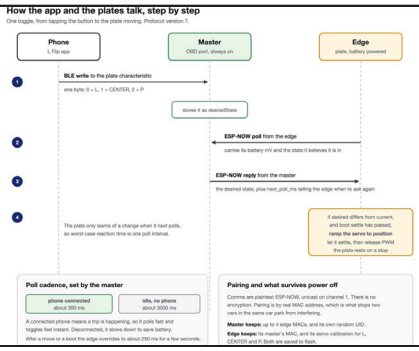
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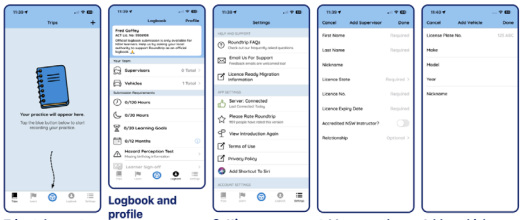
Project Development & Realisation

App Development



Existing product analysis: Roundtrip

Page 1 of 2 . Setting up the logbook



Trips tab, empty
No drives yet. An arrow points at the record button.

Logbook and profile
Licence, supervisors, vehicles, and 120 / 20 / 20 hour progress.

Settings
FAQs, server status, privacy, Siri shortcut.

Add a supervisor
Name, licence, expiry, instructor flag. All typed.

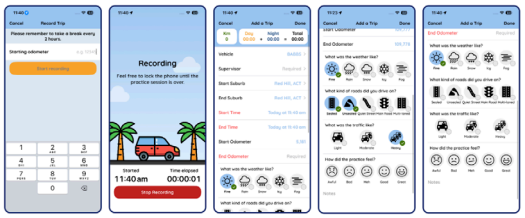
Add a vehicle
Plate, make, model, year. All typed by hand.

Analysis

What Roundtrip does well, where it falls short, and what L Flip does differently. Write here, or leave this box off and write underneath.

Existing product analysis: Roundtrip

Page 2 of 2 . Recording and entering a drive



Start a recording
Wants the start odometer first. Warns about breaks.

Recording in progress
Timer runs with the phone locked.

Add a trip, part 1
Vehicle, supervisor, suburbs, times, odometer.

Add a trip, part 2
Weather, roads, traffic, feel. Icon pickers.

Validation
Required fields turn red until filled.

Analysis

What Roundtrip does well, where it falls short, and what L Flip does differently. Write here, or leave this box off and write underneath.

How the app and the plates talk, step by step
One toggle, from tapping the button to the plate moving. Protocol version 7.
Phone
L Flip app
Master
OBD port, always on
Edge
plate, battery powered
1
BLE write to the plate characteristic
one byte: 0 = L, 1 = CENTER, 2 = P
stores it as desiredState
2
ESP-NOW poll from the edge
carries its battery mV and the state it believes it is in
3
ESP-NOW reply from the master
the desired state, plus next_poll_ms telling the edge when to ask again
4
The plate only learns of a change when it next polls, so worst case reaction time is one poll interval.
if desired differs from current, and boot settle has passed, ramp the servo to position let it settle, then release PWM the plate rests on a stop
Poll cadence, set by the master
phone connected about 300 ms
idle, no phone about 3000 ms
A connected phone means a trip is happening, so it polls fast and toggles feel instant. Disconnected, it slows down to save battery.
After a move or a boot the edge overrides to about 250 ms for a few seconds.
Pairing and what survives power off
Comms are plaintext ESP-NOW, unicast on channel 1. There is no encryption. Pairing is by real MAC address, which is what stops two cars in the same car park from interfering.
Master keeps: up to 4 edge MACs, and its own random UID.
Edge keeps: its master's MAC, and its servo calibration for L, CENTER and P. Both are saved to flash.

Project Development & Realisation

Identification and Justification of Ideas and Resources

Tools

Project Development & Realisation

Identification and Justification of Ideas and Resources

Materials

Project Development & Realisation

Identification and Justification of Ideas and Resources

Techniques

Project Development & Realisation

Use of Communication and Presentation Techniques

Project Evaluation

Evaluation of Functional and Aesthetic Aspects of Design

Project Evaluation

Evaluation of the Projects Impact on the Individual, Society and Enviornment

Project Evaluation

Relationship of the final product, system or enviornment to the project proposal